

Deepak Mallampati

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PROFESSIONAL SUMMARY

AI engineer (MS CS, 2.5+ years) designing and deploying high-performance agentic systems and production LLM inference pipelines with measurable reliability gains. Proficient across the full stack: prompt engineering, vector DB tuning, API design, and Docker-based production deployment. Delivered ~35% retrieval precision improvement, >30% hallucination reduction, and ~30% fewer post-deployment defects. Track record building scalable, low-latency AI solutions from prototype to production for enterprise clients.

CORE COMPETENCIES

Agentic Systems Design • LLM Production Inference • RAG & Vector DB Tuning • Prompt Engineering • LangChain / LlamaIndex • FastAPI / Docker / REST APIs • Python / PyTorch / HuggingFace • Multi-Provider AI Inference • scikit-learn / XGBoost • YOLOv8 / OpenCV

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer - Healthcare Technology, AI Consulting, SaaS

- Designed and deployed **high-performance agentic LLM workflows** for multi-step healthcare queries; improved agent stability by ~25%, reduced hallucinations by >30%.
- Built **RAG systems** with recursive semantic chunking, transformer embeddings, and vector DB tuning; improved retrieval precision by ~35%, grounding alignment exceeded 90%.
- Shipped **FastAPI/Flask RESTful inference APIs** containerized with Docker; load-tested for throughput; reduced post-deploy defects by ~30%.
- Shipped predictive ML pipelines (scikit-learn, XGBoost); improved recall by 15-20% on high-risk categories while holding precision >90%.
- Built preprocessing pipelines for EHR data; raised dataset reliability above 98%, cut model instability by ~40%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Applied NLP, Video Analytics

- Transformer-based hierarchical summarization over 5,000+ sessions; **60-70% review time reduction**, ~85% **highlight alignment**.
- RAG document Q&A; reduced hallucinations ~30%, raised accuracy >85%. Flask API for production.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

- SQL/T-SQL optimization; ~20% query execution reduction. Jenkins CI/CD; >30% deployment error reduction.

KEY PROJECTS

Manga Lens - Multi-Provider AI Inference Architecture

Production Chrome extension integrating four AI vision providers (Claude Sonnet, GPT-4o, GPT-4.1 Nano, Ollama local).

Multi-provider payload abstraction: WebP for cloud providers, JPEG for Ollama to avoid CUDA OOM. Seven-day translation cache, 29 sites supported. Demonstrates cross-model inference orchestration and provider-level reliability engineering.

Agentic Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline with schema-validated JSON contracts to prevent hallucination cascades. RAG-grounded CPT/ICD validation, ANN-based duplicate detection, explainable risk scoring. Production-grade agentic system with audit-ready reasoning traces.

Agentic Pixel Character Synthesis - LLM Orchestration + Generative AI

LLM-based agent orchestrator decomposes high-level prompts into Stable Diffusion + LoRA + ControlNet generation tasks. FastAPI backend, Docker deployment. Demonstrates agentic orchestration over generative AI pipeline.

Driver Drowsiness Detection - YOLOv8 Real-Time Inference

Unified YOLOv8 model; ~30% **inference latency reduction**. Sliding-window confidence aggregation for false positive suppression (~25% **reduction**). NMS tuning for stable real-time OpenCV inference.

EDUCATION

Master of Science, Computer Science - Kent State University, Ohio *Aug 2023 - May 2025*

SKILLS

Languages & Frameworks: Python, FastAPI, Flask, TypeScript, React, SQL
LLM & GenAI: RAG, agentic workflows, LangChain, LlamaIndex, prompt engineering, vector search, embeddings, structured outputs
ML & Vision: PyTorch, HuggingFace, scikit-learn, XGBoost, YOLOv8, Stable Diffusion, ControlNet
Infra: Docker, Jenkins, Grafana, CI/CD, REST APIs
Work Authorization: F-1 OPT (US-authorized now; future sponsorship required)